TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Mateusz Balcerzak	Email	mateusz.balcerzak@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer IFAM Dresden			
Address	Winterbergstraße 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	*Metal hydrides for hydrogen-related applications*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	*Energy Materials Technologies*	*New & Renewable Energy*	*Hydrogen*
Concept of Proposed Technology	*The proposed R&D project focuses on the design, development, and validation of metal hydride materials tailored for specific hydrogen-related applications defined by participating SME. The core concept is to develop metal-hydride-forming alloys or alloy families through a combination of theoretical design, simulation-based screening, targeted synthesis, and experimental validation. Depending on the selected application, the project will deliver either a novel metal hydride product or application-specific know-how that can be commercially exploited, for example through material supply, licensing, or technology transfer to metal hydride producers.* *The research will start with application-driven requirements such as operating temperature and pressure, hydrogen capacity, absorption and desorption kinetics, cycling stability, safety, scalability, and cost. Based on these requirements, thermodynamic modeling, computational materials design,*			

	and phase stability simulations will be used to identify promising alloy compositions. Selected candidates will be synthesized and experimentally characterized with respect to for example: hydrogen storage performance, selectivity, reversibility, durability, and compatibility with the intended application. Some of possible application fields include, among others, hydrogen storage, hydrogen purification and separation, hydrogen getters, hydrogen isotope separation, thermochemical hydrogen compression, and hydrides for heat storage or heat production. *In addition to material performance optimization, the project can address key challenges related to industrial deployment. These include R&D activities focused on scaling up the synthesis of metal hydrides from laboratory scale to pilot-scale production, ensuring reproducible material quality and performance. Further aspects include reduction of material cost using metal scraps or secondary raw materials, development of non-pyrophoric and inherently safe metal hydrides suitable for handling after activation, and concepts for recycling and reuse of metal hydrides, including recovery of valuable materials from hydrides at the end of life of pressure vessels. Fraunhofer IFAM will support the R&D with unique capabilities and long-standing experience in advanced metal hydrides testing including in-situ and operando characterization of metal hydrides, including neutron imaging techniques, which can be critical for understanding hydrogen transport, phase transformation, and operability of hydrides in safety-critical and highly regulated industrial applications. The project scope will be adaptable, with a starting TRL typically in the range of 2 to 3 and a targeted final TRL of 6 to 7, depending on the selected application and maturity of the material system, enabling a clear pathway toward industrial implementation and commercialization.*
Rationale for Joint R&D Project	*Metal hydrides are enabling materials for a wide range of hydrogen-related applications, including among others, hydrogen storage, purification and separation, thermochemical compression, isotope separation, getters, and thermal energy storage. As hydrogen technologies move toward industrial deployment, there is increasing demand for application-specific metal hydrides offering optimized performance, operational safety, cost efficiency, and recyclability. Many commercially available hydrides are not fully suited to these requirements, creating a clear need for targeted material development.* *Fraunhofer IFAM has extensive experience in the design, synthesis, and characterization of metal hydrides for hydrogen technologies, supported by advanced experimental infrastructure and unique in-situ and operando diagnostic capabilities. The institute possesses know-how and holds intellectual property in the field of metal hydrides, including patents related to material compositions, hydrogenation activation, distribution of metal hydride in pressure vessels and metal hydride filling sensor, which can be leveraged and further developed within the project (1-3).* *A joint R&D project between Fraunhofer IFAM and a Korean SME will combine advanced materials expertise with application-driven development and market orientation. The collaboration will enable effective transfer of metal hydride know-how and IP toward the Korean market, reduce development risks, and accelerate the creation of new metal hydride products or commercially exploitable know-how aligned with industrial needs and commercialization objectives.*

Key Words	1. *Metal hydride* 2. *Hydrogen technology* 3. *Hydrogen storage* 4. *Hydrogen separation* 5. *Recycling* 6. *Reduction of product price* 7. *Scaling up the production*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2-3)	End of the Project	Level (6-7)
R&D Objectives	*The objective of the joint R&D project is to create application-specific metal hydride materials or metal hydride families that can be commercialized as products or exploited as transferable know-how for hydrogen-related applications. The project aims to deliver new or significantly improved metal-hydride-forming alloys that meet clearly defined industrial requirements in terms of hydrogen performance, operational safety, cost efficiency, and scalability, enabling market entry or licensing by the SME.* *A key objective is to achieve validated material performance and reproducibility at a scale relevant for industrial use, demonstrating that the developed metal hydrides can be produced consistently beyond laboratory scale and are suitable for integration into real hydrogen systems. Depending on the selected application, the project will result in either a qualified metal hydride product with defined specifications or an application-specific technology package covering material composition, processing windows, and usage guidelines.* *Additional objectives include reducing material cost using optimized alloy compositions and secondary raw materials, improving safety in handling and operation through the development of non-pyrophoric hydrides, and establishing viable recycling and end-of-life concepts. The project will conclude with application-relevant validation results and a clear commercialization roadmap, supporting progression from an initial TRL of approximately 2 to 3 to a final TRL of 6 to 7.*			
R&D Target Specifications	*The project targets the development of metal hydride materials with application-specific hydrogen storage, separation, or functional performance suitable for industrial use. Depending on the selected application, the developed metal hydrides shall achieve a reversible hydrogen capacity in the range of approximately 1 to 7 wt.%, with equilibrium operating pressures tailored to the application window, typically between 0.1 and 100 bar (with the exception for hydrogen getters and metal hydrides for hydrogen compression) at operating temperatures of 0 to 150 °C (in some cases up to 400°C). Hydrogen absorption and desorption kinetics shall enable 90 percent of usable capacity to be reached within less than 30 minutes under application-relevant conditions (depending on the experiment scale and experimental geometry).* *The materials should demonstrate cycling stability with a capacity degradation over cycling of less than 20 percent, and stable performance in the presence of defined impurity levels relevant to the target application. For safety-critical uses, the project can target the development of non-pyrophoric or low-reactivity metal hydrides that can be handled in air for limited time periods after activation, with clearly defined safety characteristics.* *From an industrialization perspective, the metal hydrides shouldl be producible at kilogram-scale*			

	per batch with reproducible composition and performance, providing a clear pathway toward further scale-up. Target material cost reduction of 20 to 40 percent compared to benchmark hydrides can be addressed through optimized alloy compositions and partial use of secondary raw materials. Recycling concepts shall enable recovery of a significant fraction of valuable metals, targeting at least 70 percent material recovery at end of life. The final outcome can be validated material specifications and performance data supporting advancement from an initial TRL of 2 to 3 to a final TRL of 6 to 7, depending on application scope.
R&D Plan	*Overall approach and timeline (36 months total)* *The R&D project will be implemented over two consecutive phases, progressing from application definition and material concept selection to pilot-scale material validation and commercialization preparation. The plan is designed to ensure that material development is closely aligned with concrete industrial needs and business objectives defined by the SME.* *Example of Phase I Pre-study plan (6 months)* • *Identification and prioritization of target hydrogen-related applications together with the SME* • *Definition of application-specific material requirements including operating pressure and/or temperature, hydrogen capacity, kinetics, safety, cost, and/or recyclability targets* • *Review and benchmarking of existing metal hydride materials and competing technologies* • *Selection of promising metal hydride systems based on application requirements and theoretical considerations* • *Preliminary assessment of material production routes, cost structure, and end-of-life handling concepts* • *Initial definition of commercialization and exploitation options, including product or know-how-based business models* • *Milestone Phase I: Selection of target application(s) and metal hydride system(s), with defined material requirements and a go decision for Phase II* *Example of Phase II Main R&D (30 months)* • *Synthesis and optimization of selected metal hydride materials* • *Experimental evaluation of hydrogen performance, kinetics, cycling stability, and/or safety behavior under application-relevant conditions* • *Scale-up of material synthesis toward kilogram-level batches and assessment of reproducibility* • *Development of concepts for safe handling, recycling, and end-of-life reuse of metal hydrides* • *Advanced characterization, including in-situ and operando methods where required, to support understanding of material behavior and operability* • *Validation of material performance against target specifications and preparation of material product specifications or transferable know-how packages* • *Development of a commercialization roadmap supporting industrial deployment and market entry.* • *Milestone Phase II: Validated metal hydride material or know-how package achieving application-relevant performance and readiness for TRL 6 to 7 commercialization*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will share with SME the scientific and technological leadership of the project, including application-driven material design, synthesis at laboratory scale, and detailed characterization of metal hydride performance. The institute will co-define material target specifications together with the SME and conduct experimental validation of hydrogen capacity, kinetics, cycling stability, safety, and recyclability. Fraunhofer IFAM will provide advanced characterization capabilities, including in-situ and operando techniques such as neutron imaging where required, to support understanding of material behavior and operability. The institute will contribute its existing know-how and intellectual property in metal hydrides and support protection of new results generated during the project. Fraunhofer IFAM will also support technical risk assessment and TRL evaluation.*
Expected R&R of Korean SME Partner	*The Korean SME will lead application definition, scale-up of metal hydride synthesis toward pilot-scale production, and assessment of manufacturability and cost structure. The SME will be responsible for developing and validating scalable production routes (validation of hydrogen properties will be supported by IFAM), including use of secondary raw materials where applicable (also pretested on the lab scale), and for evaluating quality consistency at increased batch sizes. In parallel, the SME will lead business case development, market analysis, and definition of commercialization strategies, including product, licensing, or service-based models. The SME will coordinate activities related to industrial relevance, regulatory considerations, and customer requirements. Close collaboration with Fraunhofer IFAM will ensure alignment between material performance, scalable production, and commercialization objectives.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The joint R&D project is expected to deliver validated, application-specific metal hydride materials or metal hydride families that are suitable for industrial use in hydrogen-related applications such as storage, purification, compression, or thermal energy systems. The primary technical outcome will be metal hydride products or transferable know-how packages with defined specifications, proven performance, safety characteristics, and reproducibility at pilot-relevant scale, supporting advancement to TRL 6 to 7. In addition, the project will generate validated data on scalable synthesis routes, material handling, and recycling concepts, reducing technical and economic risks for industrial deployment.* *From a commercialization perspective, the project will enable the Korean SME to establish new business opportunities based on metal hydride materials, scalable production processes, or application-specific know-how. Expected outcomes include a concrete commercialization and exploitation plan covering target markets, production scale, cost structure, and revenue models such as material sales, licensing, or technology transfer. The close involvement of the SME in scale-up and business case development will significantly accelerate time to market and increase competitiveness in both Korean and international hydrogen technology markets.* *The broader impact of the project includes strengthening the domestic supply chain for advanced hydrogen materials, improving resource efficiency through recyclable and safer metal hydrides, and supporting the transition toward sustainable hydrogen technologies. The collaboration will also reinforce long-term cooperation between Korean industry and Fraunhofer IFAM, enhance innovation capacity in metal hydride technologies, and contribute to the development of safer, more cost-effective materials that enable the wider adoption of hydrogen across multiple industrial sectors.*

6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization